

CampusPilot AI - API Reference Manual

Production REST API interface definitions, payload schemas, and client examples.

1. POST /chat - Submit Conversation Message

Description: Sends a natural language query for a specific student profile. Evaluates intents, executes planned ERP tools, compiles summaries, and logs metrics in SQLite.

Request Headers

Header Key	Type	Required	Description
Content-Type	string	Yes	application/json
X-Groq-API-Key	string	No	Optional Groq key to enable Llama 3.3 agent planning.

Request Body JSON Payload

```
{
  "student_id": "ST101",
  "message": "Show my midterm marks and check pending fees."
}
```

Response JSON Payload (200 OK)

```
{
  "intent": "Multi-intent",
  "plan": [
    "Identify student ST101",
    "Load marks database",
    "Load fees database",
    "Calculate averages & outstanding dues",
    "Generate consolidated output"
  ],
  "tool": "Marks Tool, Fees Tool",
  "response": {
    "marks": {
      "student_name": "Aarav Mehta",
      "class": "10-A",
      "exams": {
        "Midterm": {
          "Mathematics": 88,
          "Science": 72
        }
      }
    },
    "fees": {
      "pending_fees": 6000,
      "status": "Pending"
    }
  },
  "summary": "You have a midterm average of 80%...",
  "status": "Pending",
  "execution_time": 1.12
}
```

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2. GET /chat/history - Retrieve Conversation Logs

Description: Fetches chronological conversation transcripts stored in SQLite.

Query Parameters

Parameter	Type	Required	Default	Description
student_id	string	No	None	Filter results specifically by student ID.
limit	int	No	50	Cap returned log list size (max 100).

Response JSON Payload (200 OK)

```
[
  {
    "id": 1,
    "student_id": "ST101",
    "query": "Show my midterm marks.",
    "intent": "Marks",
    "tool_used": "Marks Tool",
    "response": {
      "summary": "Your midterm average is..."
    },
    "timestamp": "2026-07-03T09:20:30Z",
    "execution_time": 0.85
  }
]
```

3. cURL CLI Invocation Example

```
curl -X POST "http://127.0.0.1:8000/chat" \
  -H "Content-Type: application/json" \
  -H "X-Groq-API-Key: your_key_here" \
  -d '{"student_id": "ST101", "message": "Show homework due tomorrow."}'
```